

EVALUATING CONFORMAL PREDICTION FOR UNCERTAINTY QUANTIFICATION IN MACHINE LEARNING-BASED TEMPERATURE PREDICTION

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1. INTRODUCTION

1.1. Brief introduction of the study

Machine learning (ML) has increasingly been applied in environmental research because of its ability to identify complex patterns and improve predictive performance (Bishop, 2006). Temperature prediction is one area where ML techniques have shown promise; however, many models generate point predictions without quantifying the uncertainty associated with those predictions (Murphy, 2022). This study investigates conformal prediction as a method for incorporating uncertainty quantification into machine learning-based temperature prediction, using models such as Random Forest, XGBoost, and Multilayer Perceptron (MLP).

1.2. Contextualisation

Accurate temperature predictions support decision-making in sectors such as agriculture, energy management, and environmental planning. Although ML models often achieve high predictive accuracy, uncertainty information is frequently absent from their outputs (Gal et al., 2022). Conformal prediction has emerged as a practical approach to uncertainty quantification because it provides prediction intervals with statistical coverage guarantees while remaining applicable to different ML models (Angelopoulos and Bates, 2023). Although the broader problem applies widely, this study focuses on Cape Town as a case study for evaluating the reliability of machine learning-based temperature predictions.

1.3. Rationale

As the use of ML in environmental sciences expands, there is a growing need for models that are not only accurate but also reliable. Predictions without uncertainty estimates may lead to overconfidence and limit their usefulness in decision-making (Murphy, 2022). Evaluating conformal prediction may provide insights into its suitability as an accessible uncertainty quantification technique.

1.4. Problem Statement

Many machine learning models used in temperature forecasting focus on predictive accuracy while neglecting uncertainty estimation (Gal et al., 2022). As a result, users may

be unable to assess the reliability of model outputs in contexts where forecasts support planning and decision-making. There is therefore a need to investigate practical methods, such as conformal prediction, that can provide meaningful measures of predictive uncertainty.

1.5. Research Aim

To evaluate the effectiveness of conformal prediction as an uncertainty quantification technique in machine learning-based temperature prediction.

1.6. Objectives

- To develop a machine learning model for temperature prediction.
- To apply conformal prediction to generate prediction intervals.
- To evaluate the predictive performance of the model.
- To assess the effectiveness of conformal prediction in quantifying uncertainty.

2. LITERATURE REVIEW

2.1. Introduction

Machine learning is increasingly used in weather and climate research, with temperature among one of the most studied variables and models such as Random Forest, neural networks, and XGBoost commonly applied in forecasting (Bochenek and Ustrnul, 2022; Chen et al., 2023). The literature also shows that, despite this progress, many studies still prioritise accuracy and point forecasts over predictive uncertainty (Gal et al., 2022; Murphy, 2022).

2.2. Machine Learning Methods for Temperature Forecasting

What stands out across these studies is the growing use of machine learning in temperature forecasting. Bochenek and Ustrnul (2022) identify Random Forest, artificial neural networks, support vector machines, and XGBoost as some of the most commonly used methods in weather and climate research, while Chen et al. (2023) similarly highlight the potential of machine learning in short-term weather prediction. Applied studies support this view. Sevgin (2025) found that artificial neural networks performed strongly in temperature estimation, while Random Forest was the best among the traditional machine learning models tested. Singh and Rawat (2023) also showed that forecasting performance can vary depending on the model used. For this study, the important point is that Random Forest, XGBoost, and Multilayer Perceptron (MLP) are already well established in weather-related prediction tasks, making them suitable models for further evaluation.

2.3. Need for Uncertainty Quantification in Machine Learning

A limitation in the current literature is that, although high predictive accuracy is clearly valuable, it is often treated as the main measure of model performance. Murphy (2022) explains that machine learning should ideally include a representation of uncertainty, because uncertainty is central to informed decision-making under risk. This is particularly relevant in forecasting, where users need more than a predicted value; they also need an indication of how reliable that prediction is likely to be.

Gal et al. (2022) further argue that uncertainty quantification is a key part of scientific machine learning because it shapes how much confidence can be placed in model

outputs, how easily those outputs can be interpreted, and how well they can be compared with observed data. Although the authors discuss this in relation to physics, the same concern is relevant to environmental forecasting, where changing weather conditions, noisy observations, and complex variable interactions make uncertainty unavoidable. For this study, the important point is that temperature forecasts become less useful when they provide only point predictions without indicating the likely range of the true value.

2.4. Conformal Prediction for Reliable Temperature Forecasting

One promising approach to uncertainty quantification is conformal prediction. Angelopoulos and Bates (2023) describe conformal prediction as a distribution-free and model-agnostic method that can generate statistically valid prediction intervals for machine learning models without requiring strong assumptions about data distribution. This makes it attractive in practical applications where the underlying data-generating process is difficult to specify, as is often the case in environmental data.

A major advantage of conformal prediction is that it can be applied on top of an existing predictive model rather than replacing it. In a regression problem such as temperature prediction, this means that a point forecast can be converted into a prediction interval with a chosen coverage level, such as 90%. As a result, conformal prediction does not only answer the question of what temperature is expected but also provides information about how uncertain that prediction is.

Recent applied work also suggests that conformal methods can be useful in temperature-related prediction tasks. Strickland et al. (2025) combined deep learning with conformal prediction for non-invasive core body temperature prediction and showed that the method provided calibrated uncertainty estimates and statistically valid prediction intervals in a safety-critical context. Although this study does not focus on environmental temperature, it demonstrates the practical value of conformal prediction when reliable uncertainty information is needed.

2.5. South African and Cape Town Context

South African and Western Cape studies strengthen the relevance of this research by showing that climate-related forecasting is important in a region already under environmental and resource pressure. The Western Cape has experienced recurrent drought, and evidence from station-based studies shows a long-term drying trend together with increasing warming and water deficit in the Cape Town area. Jury (2020) reports a steady drying trend around Cape Town and faster rates of warming in recent decades, while Botai et al. (2017) show that the Western Cape has experienced recurrent drought conditions with clear spatial and temporal variability.

The practical importance of forecasting in this region became especially clear during the 2015-2018 drought, which contributed to Cape Town's widely recognised "Day Zero" water crisis. Calverley and Walther (2022) show that this drought had major implications for water management and social equity, while Cash, Burls and Howar (2023) note that severe winter rainfall deficits in the Cape Town region played a central role in the crisis and were difficult to predict accurately.

The Western Cape context is also important because local agriculture is highly sensitive to climate variability. Research on the region shows that drought and changing climate conditions have affected crop production, water availability, and farm-level adaptation, particularly in high-value agricultural systems such as fruit and wine production. Studies further suggest that access to climate and weather information can improve resilience and planning in the agricultural sector.

What makes this especially relevant for the present study is that data-driven forecasting is already being used successfully in South Africa, including Cape Town, but mainly in drought-related applications. Onyeuwaoma, Sivakumar and Bade (2024) used artificial neural networks to simulate drought conditions across South African locations, including Cape Town, and found that performance depended on local meteorological factors. Similarly, Sibiya et al. (2024) used a hybrid forecasting model for Cape Town and found strong predictive performance for drought indices. Together, these studies suggest that machine learning has practical value in South African climate-related prediction, but

they also show that local applications have focused more on predictive performance than on uncertainty-aware forecasting.

2.6. Research Gap

The literature shows that machine learning has strong potential for temperature forecasting and that models such as Random Forest, XGBoost, and neural networks are already widely used. However, most existing studies focus on improving forecasting accuracy rather than evaluating whether predictions are accompanied by reliable uncertainty estimates.

At the same time, conformal prediction has a strong theoretical foundation and is presented as a practical method for uncertainty quantification across different machine learning models. Yet there are still relatively few studies that evaluate conformal prediction specifically in the context of environmental temperature forecasting, particularly in South African settings such as Cape Town. This creates a gap in the literature and justifies the present study, which aims to determine whether conformal prediction can improve the reliability and usefulness of machine learning-based temperature forecasts.

3. METHODOLOGY

3.1. Research Design

This study will adopt a quantitative research design to evaluate conformal prediction as an uncertainty quantification technique in machine learning-based temperature forecasting. A case study approach will be used, with Cape Town, South Africa, serving as the study area. This design is appropriate because the study aims to compare predictive accuracy and predictive uncertainty using measurable numerical metrics such as error, coverage probability, and interval width. Quantitative approaches are widely used in machine learning-based forecasting studies due to their reliance on data-driven modelling and objective evaluation (Chen et al., 2023; Zhang et al., 2025).

3.2. Study Area

The study will focus on Cape Town, situated in the Western Cape province of South Africa. This region is particularly relevant because it has experienced significant climate variability, including drought and warming trends. Previous studies have identified long-term drying patterns and temperature increases in the Cape Town region, as well as the severe impacts of the 2015 - 2018 drought and the “Day Zero” crisis (Jury, 2020; Botai et al., 2017; Calverley and Walther, 2022). These factors highlight the importance of accurate and reliable temperature forecasting in the local context.

3.3. Data Source

This study will use historical daily meteorological data derived from reanalysis datasets, accessed through the Open-Meteo historical weather API. Reanalysis datasets combine observational data with numerical weather prediction models to produce consistent and spatially complete representations of past atmospheric conditions. They are mostly used in climate and forecasting research because they provide continuous datasets with few or no missing values and consistent temporal coverage.

The dataset will consist of daily observations for Cape Town, South Africa, from the period 1 January 2015 to 31 December 2025, which provides sufficient data to capture seasonal patterns and interannual variability.

The variables to be used include:

- daily maximum temperature (TMAX)
- daily minimum temperature (TMIN)
- daily precipitation (PRCP)
- derived time-based variables, such as month and day of the year

These variables are commonly used in temperature forecasting studies and have been shown to provide useful predictive information (Sevgin, 2025; Chen et al., 2023).

3.4. Variables and Feature Engineering

The target variable for the study will be daily maximum temperature (TMAX). This choice is appropriate because maximum temperature is widely used in applied weather and climate prediction, and it is directly relevant to environmentally and socially important temperature extremes. Applied ML temperature studies commonly forecast maximum or minimum temperature and use related weather variables as predictors (Sevgin, 2025; Singh and Rawat, 2023).

The study will use the following predictor variables:

- daily minimum temperature (TMIN)
- daily mean temperature
- daily precipitation (PRCP)
- date-based variables, such as month and day of year
- lagged temperature variables, such as previous-day TMAX values

This selection is supported by forecasting studies showing that variables such as humidity, wind speed, precipitation, and temperature can contribute meaningfully to temperature estimation, while forecasting surveys identify temperature, precipitation, wind, pressure, and radiation as commonly used atmospheric variables in ML-based weather studies (Sevgin, 2025; Chen et al., 2023; Zhang et al., 2025).

To formulate the task as a forecasting problem, the models will predict the next day's maximum temperature. The target variable will therefore be constructed by shifting daily TMAX one day forward. Lagged variables such as TMAX lag 1, TMAX lag 2, and TMAX lag 3 will be created to capture short-term temporal dependence. Studies of seasonal and time-series forecasting show that historical values and temporal structure are important sources of predictive information in seasonal data (Kramar and Alchakov, 2023; Gong et al., 2022).

3.5. Data Preprocessing

The dataset will be prepared for modelling through the following steps:

- arranging the data in chronological order
- handling any potential inconsistencies or anomalies
- creating lagged variables
- generating date-based features
- normalising or standardising input variables where required

Because the dataset is derived from reanalysis sources, it is expected to contain few or no missing values, although data quality checks will still be performed to ensure consistency before modelling.

The dataset will be split chronologically into:

- training set (70%)
- calibration set (15%)
- test set (15%)

Chronological splitting is essential for time-series data to avoid information leakage and ensure realistic forecasting conditions.

3.6. Model Development

Three machine learning models will be developed and compared in this study:

- Random Forest (RF)
- XGBoost
- Multilayer Perceptron (MLP)

These models were selected because they represent different modelling approaches and are widely used in forecasting problems involving nonlinear relationships between variables (Bochenek and Ustrnul, 2022; Sevgin, 2025).

3.6.1. Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees during training and combines their predictions to improve accuracy and reduce overfitting.

For a regression problem, the Random Forest prediction is given by:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

where:

- b refers to the index of a single tree in the forest,
- B is the number of trees,
- $T_b(x)$ is the prediction of the b^{th} tree,
- $\hat{y}$ is the final predicted value.

Random Forest is suitable for this study because it can model nonlinear relationships, handle interactions between variables, and perform well on structured tabular data.

3.6.2. XGBoost

XGBoost (Extreme Gradient Boosting) is a boosting-based ensemble method in which models are built sequentially. Each new model attempts to correct the errors made by previous models, resulting in improved predictive performance.

The prediction for observation i is given by:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i)$$

where:

- k refers to the index of each tree (or boosting step) in the model,
- K is the number of trees,
- $f_k(x_i)$ is the contribution of the k -th tree.

XGBoost is widely used in forecasting studies due to its strong performance on structured datasets and its ability to capture complex patterns.

3.6.3. Multilayer Perceptron

The Multilayer Perceptron (MLP) is a feedforward neural network consisting of an input layer, one or more hidden layers, and an output layer.

Each neuron computes:

$$z = \sum_{j=1}^p w_j x_j + b$$
$$a = \phi(z)$$

where:

- p is the total number of input features,
- x_j are input features,
- w_j are weights,
- b is the bias term,
- $\phi(\cdot)$ is the activation function,
- a is the output of the neuron.

MLP is included because it can model complex nonlinear relationships and provides a neural network-based approach to temperature prediction.

3.6.4. Hyperparameter Tuning

Hyperparameter tuning will be applied to optimise model performance. Unlike model parameters, hyperparameters are set before training and influence how the model learns.

Examples include:

- Random Forest: number of trees, maximum depth
- XGBoost: learning rate, number of estimators, maximum depth
- MLP: number of hidden layers, neurons per layer, learning rate

Tuning will be conducted using a temporally ordered validation approach to preserve time dependencies in the data and avoid information leakage.

3.6.5. Model Assumptions and Considerations

Although these machine learning models do not rely on strict statistical assumptions such as linearity or normality, they still depend on certain practical considerations:

- The training data should be representative of future conditions
- Input variables should contain informative features
- Temporal order must be preserved to avoid leakage
- MLP requires feature scaling, which will be applied during preprocessing

These considerations ensure that the models produce reliable and meaningful predictions.

3.7. Conformal Prediction and Python Implementation

3.7.1. About Conformal Prediction

Conformal prediction is a distribution-free uncertainty quantification framework that can be applied on top of a machine learning model. In regression problems, it transforms a model's point prediction into a prediction interval that is guaranteed to contain the true value with a chosen probability level, such as 90% or 95%, under exchangeability. Angelopoulos and Bates describe conformal prediction as a model-agnostic, distribution-free framework, while MAPIE documents it as a method for computing regression

prediction intervals and time-series uncertainty estimates (Angelopoulos and Bates, 2023; MAPIE Developers, 2026a; MAPIE Developers, 2026b).

Conformal prediction is especially relevant to this study because the objective is not only to predict next-day temperature, but also to provide a meaningful measure of uncertainty around that prediction (Angelopoulos and Bates, 2023; MAPIE Developers, 2026a; MAPIE Developers, 2026b).

3.7.2. Split Conformal Prediction

The primary uncertainty quantification method in this study will be split conformal prediction. The method works by:

- training the base model on a training set
- using a separate calibration set to measure how wrong the model is
- computing nonconformity scores from the calibration errors
- using the relevant quantile of those errors to build intervals around future predictions

MAPIE's SplitConformalRegressor follows a three-step procedure: the model is fitted on the training data, conformalized on a held-out calibration set, and then used to produce prediction intervals for the test set (MAPIE Developers, 2026c; MAPIE Developers, 2026d).

For regression, a common nonconformity score is the absolute residual:

$$s_i = |y_i - \hat{y}_i|$$

where y_i is the true value and $\hat{y}_i$ is the model prediction on the calibration set. MAPIE lists "absolute" as a standard conformity-score option for split conformal regression (MAPIE Developers 2026c, 2026a).

The interval for a new input x is then expressed as:

$$[\hat{y}(x) - q_{1-\alpha}, \hat{y}(x) + q_{1-\alpha}]$$

where $q_{1-\alpha}$ is the $(1 - \alpha)$ -quantile of the calibration nonconformity scores. This split-conformal workflow is described in the conformal prediction literature and is implemented

in MAPIE’s regression API through *SplitConformalRegressor* (Angelopoulos and Bates, 2023; MAPIE Developers, 2026a; MAPIE Developers, 2026b).

3.7.3. Exchangeability and Time-Series Data

The key assumption behind standard conformal prediction is exchangeability. In simple terms, this means that the calibration data and future test data are generated in a sufficiently similar way for calibration errors to remain informative for future predictions. *MAPIE* and *PUNCC* both describe conformal prediction’s finite-sample coverage guarantee under exchangeability or conditions resembling independent and identically distributed (i.i.d.) data (MAPIE Developers 2026a; PUNCC Developers 2026a, 2026b).

Because this study uses time-series temperature data, strict exchangeability is more difficult to justify if observations are randomly shuffled. For this reason, the study will preserve temporal order and use chronological train–calibration–test splits. The calibration set will therefore be a later block of historical data rather than a random sample. MAPIE’s *TimeSeriesRegressor* and *BlockBootstrap* utilities reflect the need to handle time-dependent data more carefully than ordinary independent and identically distributed regression settings. Since the study uses a single consistent reanalysis dataset, preserving temporal order helps keep calibration and test conditions as comparable as possible (MAPIE Developers 2026c, 2026b; Open-Meteo 2026b, 2026a).

3.7.4. Python Libraries for Conformal Prediction

The primary Python library that will be used for conformal prediction in this study is *MAPIE* (Model Agnostic Prediction Interval Estimator). *MAPIE* was selected because it is *scikit – learn*-compatible, supports regression and time-series prediction intervals, and includes built-in evaluation metrics such as coverage probability and mean interval width. MAPIE’s regression conformalizers include *SplitConformalRegressor*, *CrossConformalRegressor*, *JackknifeAfterBootstrapRegressor*, *ConformalizedQuantileRegressor*, and *TimeSeriesRegressor*, which makes it particularly suitable for the workflow proposed in this study (MAPIE Developers 2026b, 2026c, 2026e).

Alternative conformal prediction libraries such as *PUNCC*, *TorchCP*, and *crepes* were also considered. *PUNCC* provides conformal methods for regression, classification, anomaly detection, and time-series workflows; *TorchCP* is more oriented toward *PyTorch*-heavy deep-learning research; and *crepes* supports standard, normalized, and Mondrian conformal methods. However, *MAPIE* is the most appropriate choice for this study because it aligns best with the selected model types and the planned regression workflow (PUNCC Developers 2026a, 2026b; TorchCP Developers 2026; Huang et al. 2025; crepes Developers 2026; Boström 2022).

3.7.5. Software and Planned Implementation

The study will use the following Python libraries:

- pandas – data handling and preprocessing
- numpy – numerical operations
- scikit-learn – Random Forest, MLP, preprocessing, and evaluation
- xgboost – XGBoost modelling
- mapie – conformal prediction and interval evaluation

These packages support a transparent and reproducible modelling workflow. *MAPIE* is designed to wrap scikit-learn-compatible models for conformal prediction, and the package documentation shows a regression workflow based on fit, conformalize, and predict_interval (MAPIE Developers 2026b, 2026c, 2026d).

3.8. Evaluation Metrics

3.8.1. Predictive Performance

The predictive accuracy of the machine learning models will be assessed using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

These metrics are standard in temperature forecasting studies and are commonly used for comparing model performance in regression problems (Singh and Rawat 2023; Sevgin 2025).

3.8.2. Uncertainty Performance

The effectiveness of conformal prediction will be assessed using:

- Coverage Probability: the proportion of observed values that fall within the prediction intervals
- Average Interval Width: the mean width of the prediction intervals
- Interval Efficiency: the balance between achieving reliable coverage and keeping intervals sufficiently narrow

MAPIE's regression metrics include *regression_coverage_score* and *regression_mean_width_score*, which support this evaluation directly (MAPIE Developers 2026c).

4. RESULTS AND DISCUSSION

This section presents and discusses the results obtained from the exploratory data analysis, machine learning model evaluation, hyperparameter tuning, conformal prediction, and baseline comparison. The purpose of the analysis was to evaluate whether conformal prediction can provide useful uncertainty estimates for machine learning-based next-day maximum temperature forecasting in Cape Town.

4.1. Exploratory Data Analysis

The dataset consists of 4,018 daily observations from 1 January 2015 to 31 December 2025. The original dataset contains no missing values or duplicate records, indicating that the reanalysis data provided a complete and consistent time series for modelling. The summary statistics shows that daily maximum temperature ranges from 10.9°C to 37.4°C, with a mean of 22.23°C and a median of 22.05°C. The closeness of the mean and median suggests that the distribution of maximum temperature is reasonably balanced around the centre. Most daily maximum temperature values are concentrated between 18.1°C and 25.6°C, based on the interquartile range. This indicates that the temperature values were reasonably centred around 22°C, with some variation across cooler and warmer days.

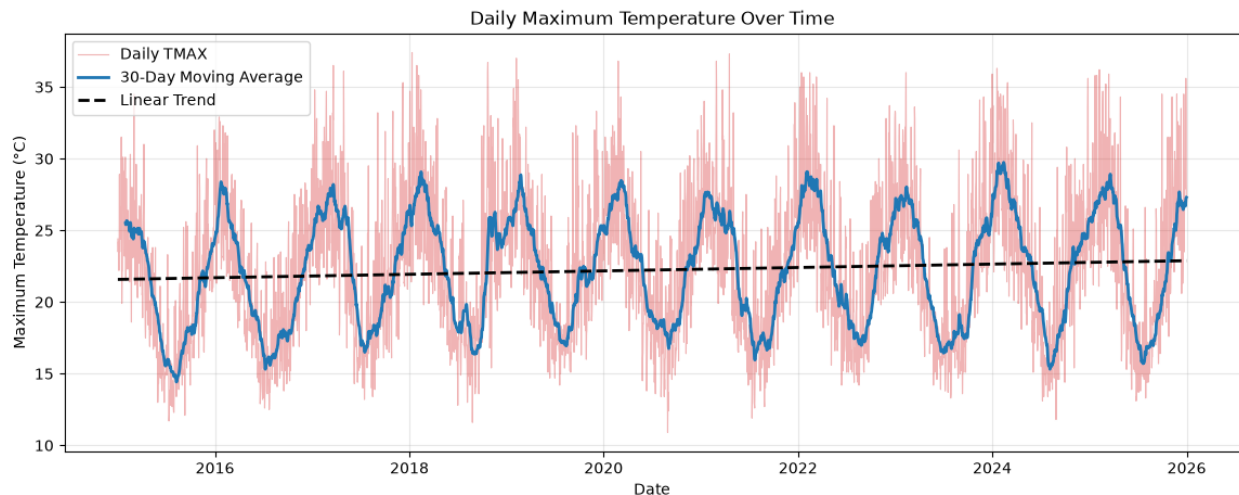


Figure 4-1: Daily Maximum Temperature Over Time

Figure 4.1 shows strong day-to-day variation, which reflects normal weather variability. The 30-day moving average smooths these short-term fluctuations and makes the

broader pattern easier to observe. The repeated rises and falls in the moving average indicate clear seasonal behaviour in maximum temperature across the study period. The linear trend line provides a general indication of long-term direction, but the most visible pattern is the recurring seasonal cycle. This supports the inclusion of lagged temperature variables and cyclical time-based features in the machine learning models.

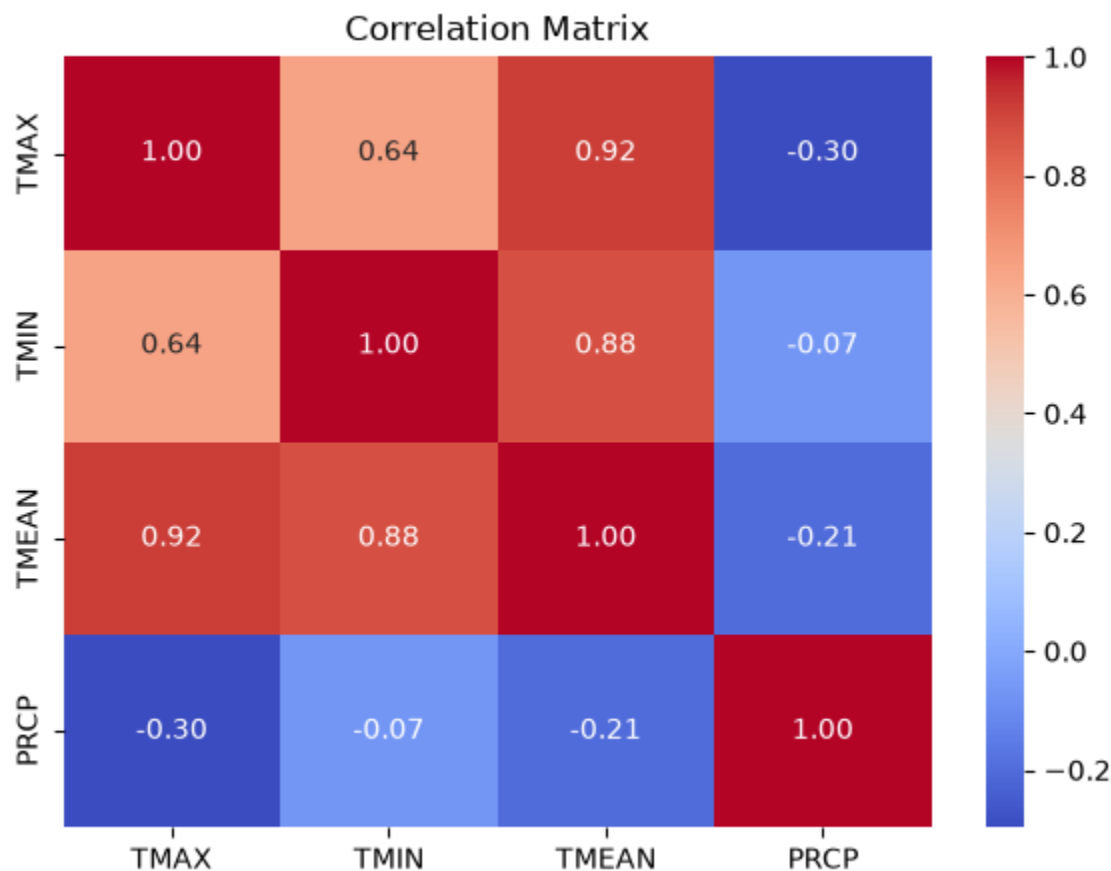


Figure 4-2: Correlation matrix

The correlation analysis shows that daily mean temperature (TMEAN) has the strongest positive relationship with daily maximum temperature (TMAX), with a correlation of approximately 0.92. Daily minimum temperature (TMIN) also has a moderate positive relationship with TMAX, with a correlation of approximately 0.64. Precipitation (PRCP) has a weak negative relationship with TMAX, with a correlation of approximately -0.30. This suggests that wetter days may be associated with slightly lower maximum temperatures. However, PRCP is retained as a predictor because weak linear correlation does not necessarily mean that a variable has no predictive value, especially in nonlinear

machine learning models that can capture more complex relationships and interactions between variables.

4.2. Feature Engineering and Data Splitting

The forecasting target is created by shifting daily maximum temperature one day forward, producing TMAX_next as the next-day maximum temperature. Lagged variables, namely TMAX_lag1, TMAX_lag2, and TMAX_lag3, are included to capture short-term temporal dependence. Cyclical encoding was also applied to month and day-of-year variables using sine and cosine transformations, allowing the models to better represent seasonal patterns.

After feature engineering, the modelling dataset contains 4,014 observations, reduced from 4,018 because the lag variables remove the first three rows and the next-day target removes the final row. The data are then split chronologically into training, calibration, and test sets containing 2,809, 602, and 603 observations, respectively. This chronological split preserves the time order of the data and helps avoid information leakage.

4.3. Initial Model Performance Before Hyperparameter Tuning

Three machine learning models were first evaluated before hyperparameter tuning: Random Forest, XGBoost, and Multilayer Perceptron (MLP). The models were compared using MAE, RMSE, and R^2 .

The initial model results are shown below:

Model	MAE	RMSE	R^2	Coverage	Average Interval Width
Random Forest	2.363	3.097	0.654	0.892	10.185
XGBoost	2.361	3.123	0.649	0.904	10.586
MLP	2.273	3.074	0.660	0.902	10.290

Before hyperparameter tuning, the MLP model performs best overall. It achieves the lowest MAE of 2.273°C, the lowest RMSE of 3.074°C, and the highest R^2 value of 0.660,

indicating that it produces the most accurate next-day maximum temperature predictions among the initial models.

Random Forest and XGBoost show similar predictive performance, although Random Forest performs slightly better in terms of RMSE and R^2 . In terms of uncertainty performance, XGBoost achieves the highest coverage at 90.38%, slightly above the nominal 90% level, but it also produces the widest prediction intervals. MLP achieves coverage of 90.22% while maintaining the best predictive accuracy. This suggests that, before tuning, MLP provides the strongest overall balance between predictive accuracy and uncertainty performance.

4.4. Hyperparameter Tuning Results

Hyperparameter tuning was performed using *GridSearchCV* with *TimeSeriesSplit*. *TimeSeriesSplit* was used because the data are time-dependent, and random cross-validation would not be appropriate for a forecasting problem. MAE was used as the tuning criterion because it is directly interpretable as average temperature prediction error in degrees Celsius.

The hyperparameter tuning results are summarised below:

Model	Best CV MAE
Random Forest	2.320
XGBoost	2.300
MLP	2.362

During cross-validation on the training set, XGBoost records the lowest validation MAE, followed by Random Forest and MLP. This suggests that XGBoost performs best during hyperparameter tuning. However, final model selection is based on the independent test set after conformal prediction is applied. This is important in time-series forecasting because the final test period represents later observations that may differ from the training and validation periods.

4.5. Tuned Model Performance

After hyperparameter tuning, the models were evaluated again on the test set and conformal prediction was applied to each tuned model.

The tuned results are shown below:

Model	MAE	RMSE	R ²	Coverage	Average Interval Width
Random Forest	2.327	3.069	0.661	0.886	9.924
XGBoost	2.313	3.089	0.656	0.884	9.682
MLP	2.268	3.038	0.667	0.889	9.611

After hyperparameter tuning, all three models show modest improvements in predictive performance. MAE and RMSE decrease, while R² increases across the models, indicating that tuning improves point prediction accuracy, although the gains are relatively small.

The tuned MLP model remains the best-performing model overall. It achieves the lowest MAE of 2.268°C, the lowest RMSE of 3.038°C, and the highest R² value of **0.667**. This indicates that MLP produces the most accurate next-day maximum temperature forecasts among the tuned models. In practical terms, the tuned MLP predicts next-day maximum temperature with an average error of approximately 2.27°C.

Random Forest also improves after tuning, with MAE decreasing from 2.363°C to 2.327°C and R² increasing from 0.654 to 0.661. XGBoost also improves in terms of MAE and interval width, but it does not outperform MLP on the final test set.

In terms of uncertainty performance, the tuned MLP produces the narrowest average prediction intervals, with an interval width of 9.611°C. This suggests that MLP provides the most efficient uncertainty estimates among the tuned models, while also maintaining the strongest predictive accuracy.

4.6. Conformal Prediction Performance

The main purpose of this study was not only to evaluate predictive accuracy, but also to assess whether conformal prediction provides useful uncertainty estimates. The

conformal confidence level is set at 90%, meaning that the prediction intervals are expected to contain the true temperature values approximately 90% of the time.

Before hyperparameter tuning, XGBoost and MLP achieve coverage slightly above the nominal level, with coverage values of 90.38% and 90.22%, respectively. This indicates that split conformal prediction produces prediction intervals with empirical coverage close to the target level.

After tuning, the coverage values decrease slightly below 90%. The tuned Random Forest model achieves coverage of 88.56%, XGBoost achieves 88.39%, and MLP achieves 88.89%. Although these values are slightly below the nominal level, they remain close to the target coverage.

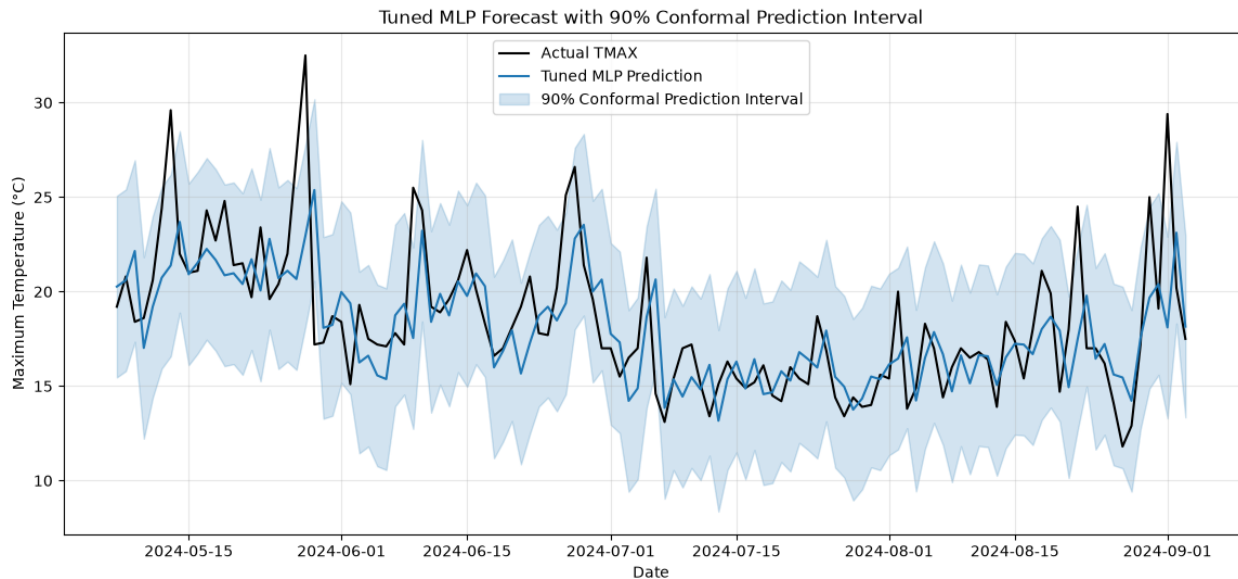


Figure 4.6: Tuned MLP Forecast with 90% Conformal Prediction Interval

Figure 4.6 presents the tuned MLP forecasts with 90% conformal prediction intervals. The actual TMAX values mostly fall within the shaded interval, showing that conformal prediction provides useful uncertainty estimates around the point forecasts. Some sharp temperature changes fall outside the interval, which is consistent with the slightly below-90% empirical coverage observed in the results.

The decrease in coverage after tuning reflects a trade-off between interval width and coverage. Hyperparameter tuning improves point prediction accuracy and reduces the

size of prediction errors, which leads to narrower conformal intervals. While narrower intervals are more efficient and easier to interpret, they may also miss slightly more true values. This explains why the tuned models produce narrower intervals but slightly lower empirical coverage.

This result does not suggest that conformal prediction fails. Instead, it shows that the tuned models produce sharper intervals while slightly under-covering the target level. The slight undercoverage may also be influenced by the time-series nature of the data, since temperature observations are temporally dependent and seasonal. This makes the standard exchangeability assumption of split conformal prediction more difficult to satisfy fully.

Overall, the conformal prediction results suggest that the method provides useful uncertainty estimates for machine learning-based temperature forecasting. The prediction intervals achieve empirical coverage close to the target level and provide additional information about forecast reliability beyond point predictions alone.

4.7. Overall Model Comparison

The results show that model performance depends on the evaluation criterion considered. In terms of predictive accuracy, the tuned MLP model performs best, as it achieves the lowest MAE and RMSE and the highest R^2 value among the tuned models.

From an uncertainty perspective, the tuned MLP also produces the narrowest average prediction intervals. Although its coverage is slightly below the nominal 90% level, it remains the closest to the target among the tuned models. This suggests that MLP provides the best balance between predictive accuracy and interval efficiency.

XGBoost performs well during hyperparameter tuning and records the lowest cross-validation MAE. However, it does not generalise as well as MLP on the final test set. Random Forest also shows reasonable performance and improves after tuning, but it remains weaker than MLP overall.

Therefore, the tuned MLP model is selected as the best overall model in this analysis.

4.8. Discussion in Relation to the Research Aim

The aim of this study was to evaluate the effectiveness of conformal prediction as an uncertainty quantification technique in machine learning-based temperature prediction. The results show that conformal prediction was useful because it transformed point predictions into prediction intervals. This directly addresses the problem that standard machine learning forecasts often provide only single-value predictions without indicating uncertainty. By producing prediction intervals, the models provide additional information about the reliability of each forecast.

Among the models evaluated, the tuned MLP model was the strongest overall model, achieving the best predictive accuracy and the narrowest prediction intervals. The conformal prediction results were close to the target 90% coverage level, although the tuned models slightly undercovered. This suggests that conformal prediction was effective, but not perfect. The slight undercoverage may reflect the challenges of applying standard conformal prediction to time-series data, where the exchangeability assumption is harder to satisfy because of temporal dependence and seasonality.

Overall, the findings support the use of conformal prediction as a practical uncertainty quantification method for machine learning-based temperature forecasting. The results show that uncertainty-aware forecasting is more informative than relying on point predictions alone.

4.9. Limitations of the Results and Future Directions

Although the results are meaningful, several limitations should be noted. First, the study uses a limited set of meteorological predictors, mainly temperature and precipitation variables. Including additional atmospheric variables such as humidity, wind speed, solar radiation, cloud cover, and atmospheric pressure may improve forecasting performance.

Second, the conformal prediction intervals remain relatively wide, with average widths of approximately 9.6°C to 9.9°C after hyperparameter tuning. This indicates that daily maximum temperature forecasting still contains considerable uncertainty. However, these intervals provide useful information about forecast reliability that would not be available from point predictions alone.

Third, the tuned models slightly under-cover the nominal 90% confidence level. This may be due to the time-series nature of the data, where observations are temporally dependent and seasonal. As a result, the exchangeability assumption required for standard split conformal prediction is more difficult to satisfy fully.

Finally, although hyperparameter tuning improves model performance, the improvements are modest and do not change the overall ranking of the models. Future work may consider more extensive hyperparameter tuning, additional meteorological predictors, alternative forecasting horizons, or time-series-specific conformal prediction methods.

5. CONCLUSION

This study evaluates conformal prediction as an uncertainty quantification technique for machine learning-based next-day maximum temperature forecasting in Cape Town. The results show that conformal prediction adds value by converting point forecasts into prediction intervals, providing useful information about forecast uncertainty.

The tuned MLP model performs best overall, achieving the lowest prediction errors, highest R^2 value, and narrowest conformal prediction intervals. Although the empirical coverage is slightly below the nominal 90% level, it remains close to the target, suggesting that conformal prediction provides reasonably reliable uncertainty estimates.

Overall, the findings support the use of conformal prediction for uncertainty-aware temperature forecasting. Future work should consider additional meteorological predictors, time-series specific conformal methods, and longer forecasting horizons to improve both predictive accuracy and interval reliability.

6. REFERENCE LIST

- Angelopoulos, A.N. and Bates, S., 2023. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4), pp.494-591.
- Bishop, C.M. and Nasrabadi, N.M., 2006. *Pattern recognition and machine learning*. New York: springer.
- Bochenek, B. and Ustrnul, Z., 2022. Machine learning in weather prediction and climate analyses - Applications and perspectives. *Atmosphere*, 13(2), p.180.
- Boström, H., 2022. crepes: A Python package for generating conformal regressors and predictive systems. *Proceedings of Machine Learning Research*, 179, pp.1–18
- Botai, C.M., Botai, J.O., De Wit, J.P., Ncongwane, K.P. and Adeola, A.M., 2017. Drought characteristics over the Western Cape Province, South Africa. *Water*, 9(11), p.876.
- Calverley, C.M. and Walther, S.C., 2022. Drought, water management, and social equity: Analysing Cape Town's water crisis. *Frontiers in Water*, 4.
- Cash, B.A., Burls, N.J. and Howar, L.V., 2023. Predictable and unpredictable components of Cape Town winter rainfall. *Journal of Climate*, 36(16), pp.5351-5362.
- Chen, L., Han, B., Wang, X., Zhao, J., Yang, W. and Yang, Z., 2023. Machine learning methods in weather and climate applications: A survey. *Applied Sciences*, 13(21), p.12019.
- crepes Developers, 2026. *crepes documentation*. <https://crepes.readthedocs.io/en/stable/> [20 June 2026].
- Gal, Y., Koumoutsakos, P., Lanusse, F., Louppe, G. and Papadimitriou, C., 2022. Bayesian uncertainty quantification for machine-learned models in physics. *Nature Reviews Physics*, 4(9), pp.573-577.
- Gong, B., Langguth, M., Ji, Y., Mozaffari, A., Stadtler, S., Mache, K. and Schultz, M.G., 2022. Temperature forecasting by deep learning methods. *Geoscientific Model Development*, 15, pp.8931–8956.

Huang, J., Song, J., Zhou, X., Jing, B. and Wei, H., 2025. TorchCP: A Python library for conformal prediction. *Journal of Machine Learning Research*, 26, pp.1–25.

Jury, M.R., 2020. Climate trends in the Cape Town area, South Africa. *Water SA*, 46(3), pp.438-447.

Kramar, V. and Alchakov, V., 2023. Time-series forecasting of seasonal data using machine learning methods. *Algorithms*, 16(5), p.248.

MAPIE Developers, 2026a. *Conformal prediction framework*. <https://deepwiki.com/scikit-learn-contrib/MAPIE/2.1-conformal-prediction-framework> [20 June 2026].

MAPIE Developers, 2026b. *MAPIE — Model Agnostic Prediction Interval Estimator*. <https://mapie.readthedocs.io/> [20 June 2026].

MAPIE Developers, 2026c. *MAPIE regression API*. <https://contrib.scikit-learn.org/MAPIE/stable/api/regression/> [20 June 2026].

MAPIE Developers, 2026d. *Quick start with MAPIE*. <https://contrib.scikit-learn.org/MAPIE/latest/getting-started/quick-start/> [20 June 2026].

Murphy, K.P., 2022. *Probabilistic machine learning: An introduction*. Cambridge, MA: MIT Press.

Onyeuwaoma, N., Sivakumar, V. and Bade, M., 2024. Modelling drought in South Africa: meteorological insights and predictive parameters. *Environmental Monitoring and Assessment*, 196(10), p.965.

Open-Meteo, 2026a. *Free open-source weather API*. <https://open-meteo.com/> [21 June 2026].

Open-Meteo, 2026b. *Historical weather API*. <https://open-meteo.com/en/docs/historical-weather-api> [21 June 2026].

PUNCC Developers, 2026a. *PUNCC documentation*. <https://deel-ai.github.io/puncc/> [21 June 2026].

PUNCC Developers, 2026b. *PUNCC on PyPI*. <https://pypi.org/project/puncc/> [21 June 2026].

Sevgin, F., 2025. Machine learning-based temperature forecasting for sustainable climate change adaptation and mitigation. *Sustainability*, 17(5), p.1812.

Sibiya, S., Mbatha, N., Ramroop, S., Melesse, S. and Silwimba, F., 2024. Forecasting of standardized precipitation index using hybrid models: A case study of Cape Town, South Africa. *Water*, 16(17), p.2469.

Singh, D.K. and Rawat, N., 2023. Machine learning for weather forecasting: XGBoost vs SVM vs Random Forest in predicting temperature for Visakhapatnam. *International Journal of Intelligent Systems and Applications*, 15(5), pp.57-69.

Strickland, J., Ghisoni, M., Marshall, H., Whitehead, T., Nenchev, B., Pellegrini, B., Phillips, C., Tassenberg, K., Davey, S., Dorman, S., Sol, J., Ferguson, D. and Conduit, G., 2025. Degrees of uncertainty: conformal deep learning for non-invasive core body temperature prediction in extreme environments. *Communications Engineering*, 4, p.219.

Theron, S.N., Midgley, S., Hochrainer-Stigler, S., Archer, E., Tramberend, S. and Walker, S., 2023. Agricultural resilience and adaptive capacity during severe drought in the Western Cape, South Africa. *Regional Environmental Change*, 23(3), p.98.

TorchCP Developers, 2026. *TorchCP documentation*. <https://torchcp.readthedocs.io/en/latest/> [23 June 2026].

Zhang, H., Liu, Y., Zhang, C. and Li, N., 2025. Machine learning methods for weather forecasting: A survey. *Atmosphere*, 16(1), p.82.

Zwane, E.M., 2019. Impact of climate change on primary agriculture, water sources and food security in Western Cape, South Africa. *Jàmbá: Journal of Disaster Risk Studies*, 11(1).